

Seongkyu CHOI

M.S. in Artificial Intelligence | Gachon University

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EDUCATION

Gachon University, Seongnam, South Korea (Mar. 2024 – Aug. 2026)

M.S. in Artificial Intelligence

- Department of AI & Software
- Graduate School, Master's Program
- Research Area: Computer Vision, Autonomous Driving, Off-Road, Multi-Task Learning, Edge AI Devices

Hansei University, Gunpo, South Korea (Mar. 2019 - Feb. 2023)

B.S. in Department of International Business

- Department of International Business

Shingu College, Seongnam, South Korea (Mar. 2015 - Feb. 2019)

Associate Degree in Business Administration

- Department of Global Business
- Transferred to Hansei University

RESEARCH INTERESTS

- Semantic segmentation in LiDAR point clouds and utilizing terrain information.
 - Computer Vision for Autonomous Driving and Robotics
 - Multi-Task Perception and Scene Understanding
 - Off-Road Semantic Segmentation under Ambiguous Boundaries and Noisy Supervision
 - Efficient On-Board AI Model Optimization for Edge Devices
 - Multi-Modal Perception using Camera, LiDAR, and Thermal Sensors
 - 3D Occupancy Prediction and Terrain Understanding

WORK EXPERIENCE

Research Experience

- Research Student, Vehicle Intelligence Perception Lab, Gachon University, Korea | Mar. 2024 ~ Aug. 2026
 - Developed off-road semantic segmentation methods using cross-scale decoding, token refinement, and transition-aware refinement.
 - Collected data with project teams and built training datasets through annotation, augmentation, and quality review.
 - Developed perception models using camera–LiDAR knowledge distillation and multi-task learning for traversability estimation and vehicle perception.
 - Optimized perception and vision-language models for Jetson deployment using TensorRT and quantization, evaluating latency, throughput, and memory usage.
 - Integrated perception pipelines using ROS2 and Docker and evaluated their applicability to real-world patrol-robot demonstration settings.

Work Experience

- Engineer, FNS, Korea | Jun. 2021–Mar. 2023
 - Performed maintenance, troubleshooting and recovery of semiconductor clean-process equipment and provided on-

site technical support at customer manufacturing sites.

- Conducted pre-shipment and post-shipment equipment inspections to verify operational stability and quality requirements.
- Analyzed recurring equipment issues and supported preventive actions to improve operational reliability.

PROJECTS

- **Development of Wide Semantic Map Building and Update Technology for Multi-Robot Systems in Unstructured Outdoor Environments**
 - MSS-funded National R&D Project led by Mobiltech
 - Project period: Apr. 2025–Mar. 2027 | Participation: Apr. 2025–Aug. 2026
 - Role: Off-road perception and traversability model development
 - ◆ Developed off-road semantic segmentation models using RGB and LiDAR information, transferring image-derived knowledge to LiDAR-based learning through knowledge distillation.
 - ◆ Converted semantic predictions into terrain-specific traversability levels and contributed to path-similarity recognition and multi-scan change detection for semantic map updates.
 - ◆ Recorded first-stage mIoU scores of 49.3 for traversable-area recognition and 47.6 for semantic segmentation.
- **Development of Intelligent Battlefield Situational Awareness AI Algorithms**
 - Industry-commissioned Research Project | Hyundai Rotem
 - Project period: Nov. 2024–Sep. 2026 | Participation: Nov. 2024–Aug. 2026
 - Role: RGB/thermal dataset development and C++ inference deployment
 - ◆ Built RGB/thermal datasets through collection, augmentation, quality review, and semi-automatic labeling, expanding the training dataset from approximately 1,500 to 4,000 images.
 - ◆ Deployed a C++ inference pipeline using ONNX and TensorRT, reducing inference latency from 28.29 to 14.18 ms while maintaining mAP50 of 0.595 and mAP50–95 of 0.354.
 - ◆ Verified simultaneous processing of two camera streams in the integrated C++ perception pipeline.
- **Development of an Automotive Software Platform for Software-Defined Vehicles (SDV) Integrated with an AI Framework Required for Intelligent Vehicles**
 - IITP-funded National R&D Project led by HL Mando
 - Project period: Apr. 2024–Dec. 2027 | Participation: Apr. 2024–Aug. 2026
 - Role: Vehicle perception development and on-board optimization
 - ◆ Developed and validated object detection, lane-line perception, and drivable-area segmentation on Jetson AGX Orin. Simplified lane-model refinement by reducing it from three stages to two and sharing refinement heads.
 - ◆ Profiled the processing pipeline, identified visualization as the main source of delay within the output stage, and prioritized rendering optimization.
 - ◆ Improved end-to-end perception-pipeline throughput from 3.73 to 24.56 FPS through OpenCV-based visualization optimization, a three-stage asynchronous pipeline, and TensorRT FP16 acceleration.
- **Development of a Network for Determining Traversable Regions by Terrain**
 - Industry-commissioned Research Project | Mobiltech
 - Period: Sep. 2024–Feb. 2025
 - Role: Camera-only traversability modeling and on-device deployment
 - ◆ Developed a traversability model for agricultural environments using LiDAR knowledge distillation, enabling camera-only inference without LiDAR input.
 - ◆ Achieved 0.87 mIoU for semantic segmentation and 0.91 mIoU for traversability classification.

- ◆ Implemented camera-only segmentation and distance regression on Jetson Orin Nano at ~ 0.1 s/frame.
- **Development of Camera-Based Patrol Cognitive Intelligence for Abnormal Situation Detection**
 - Industry-commissioned Research Project | Neubility
 - Period: Nov. 2024–Mar. 2025
 - Role: Dataset development, VLM compression, and robot perception integration
 - ◆ Collected robot-view scenario data with the project team and performed quality review and annotation to build training datasets for patrol-object and abnormal-situation recognition.
 - ◆ Applied AWQ quantization to VILA1.5-3B, reducing model size from 10.06 to 1.57 GB for memory-constrained on-board deployment.
 - ◆ Integrated YOLO detection with vision-language model descriptions on Jetson Orin Nano using ROS2 and Docker and evaluated applicability to real-world patrol-robot demonstration settings.
- **Development of Multi-Sensor Fusion Algorithm for Countering Magnetic/Acoustic Stealth Targets**
 - Industry-commissioned Research Project | LIG Nex1
 - Period: Apr. 2024–Mar. 2025
 - Role: Sea-trial data quality validation and model-result analysis
 - ◆ Screened 46 company-provided sea-trial data files from the second delivery for collection errors and recording gaps.
 - ◆ Selected 23 usable files for subsequent analysis based on the data-quality review.
 - ◆ Analyzed training and evaluation results of models developed by the project team, comparing predictive performance and assessing sensor-data suitability for training.

PUBLICATIONS

International Journal Articles

- **Choi, Seongkyu**; An, Jhonghyun. “Enhancing Group Attention for Off-Road Semantic Segmentation via Transition-Aware Refinement,” IEEE Signal Processing Letters, vol. 33, pp. 2535–2539, 2026.
- **Choi, Seongkyu**; An, Jhonghyun. "A Cross-Scale Decoder with Token Refinement for Off-Road Semantic Segmentation." Applied Sciences, vol. 16, no. 11, Art. no. 5238, 2026.
- Kim, Nahyeong; **Choi, Seongkyu**; Choi, Sun; Lee, Yejun; Cheong, Youngjae; An, Jhonghyun. “GPT-4off: On-Board Traversability Probability Estimation for Off-Road Driving via GPT Knowledge Distillation,” Applied Sciences, vol. 15, no. 4, Art. no. 2130, Feb. 2025.
(Equal contribution.)

RELATED COURSES

- Artificial Intelligence | B0
- Optimization Theory for Machine Learning | B0
- Advanced Topics in AI Autonomous Driving | A0
- Internet and Mobile Computing | A0
- Fashion Big Data Analysis and Application | A+
- Understanding of Advanced Artificial Intelligence | A+
- Probabilistic Robotics | A+
- Intelligent Vehicle Technology | A+

AWARDS

- **President’s Award**, Graduate School, Gachon University | Aug. 2026
- **Outstanding Poster Paper Award**, Korean Society of Next-generation Computing Conference | Apr. 2024
 - Paper: “Real-time Performance Optimization of AI Models Using Edge Computing”

Technical Skills

Programming

Python 
C++ 

Frameworks

PyTorch 
TensorFlow 
OpenMMLab 

Tools & Platforms

Computer Vision: OpenCV
Deployment & Optimization: ONNX, TensorRT, AWQ quantization
Robotics: ROS2
Edge Platforms: Jetson AGX Orin, Jetson Orin Nano
Containers & Cloud: Docker, AWS
Development Tools: Git, GitHub, VS Code, Anaconda
OS: Linux, Windows, macOS